

CSCE 763: Digital Image Processing

Homework #1

Instructor: Spring 2026
University of South Carolina
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Due: 1:15 pm EST, Tuesday, Feb 3

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Problem 1: Smallest Discernible Dot (25 pts)

Thinking purely in geometric terms, estimate the diameter of the smallest printed dot that the eye can discern if the page on which the dot is printed is 0.5 m away from the eyes.

Assumptions:

1. The distance between the center of the lens and the retina along the visual axis is 14 mm.
2. The visual system ceases to detect the dot when the image of the dot on the fovea becomes smaller than the diameter of one receptor (cone) in that area of the retina.
3. The fovea can be modeled as a square array of dimensions $1.5 \text{ mm} \times 1.5 \text{ mm}$, and the cones (about 337,000 in total) and spaces between the cones are distributed uniformly throughout this array.

Step 1: Determine the Diameter of a Single Cone

The fovea is a $1.5 \text{ mm} \times 1.5 \text{ mm}$ square with 337,000 cones uniformly distributed. Modeling as a square grid, the number of cones per side is:

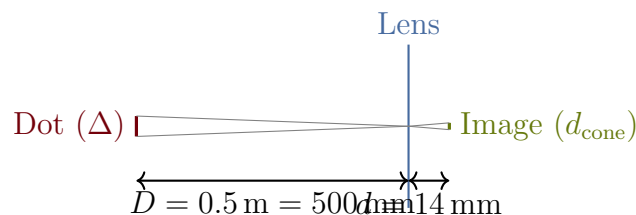
$$n = \sqrt{337,000} \approx 580.5$$

The center-to-center spacing (cone diameter including gaps) is:

$$d_{\text{cone}} = \frac{1.5 \text{ mm}}{580.5} \approx 0.00258 \text{ mm} = 2.58 \mu\text{m}$$

Step 2: Apply Similar Triangles

The dot on the page and its image on the retina are related by similar triangles through the lens center:



By similar triangles:

$$\frac{\Delta}{D} = \frac{d_{\text{cone}}}{d}$$

Step 3: Solve for Minimum Dot Diameter

The smallest detectable image equals one cone diameter: $d_{\text{image}} = d_{\text{cone}} \approx 0.00258 \text{ mm}$.

$$\Delta = D \cdot \frac{d_{\text{cone}}}{d} = 500 \cdot \frac{0.00258}{14} = \frac{1.29}{14} \approx 0.0922 \text{ mm}$$

Results

The diameter of the smallest printed dot that the eye can discern at a distance of 0.5 m is:

$$\Delta \approx 0.0922 \text{ mm} \approx 92.2 \mu\text{m}$$

Problem 2: Adjacency of Image Subsets (25 pts)

Consider the two image subsets S_1 and S_2 shown below. For $V = \{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent.

	S_1					S_2				
0	0	0	0	0	0	0	0	1	1	0
1	0	0	1	1	0	1	0	0	0	1
1	0	0	1	0	1	1	0	0	0	0
0	0	1	1	0	0	0	0	0	0	0
0	0	1	1	1	0	0	1	1	1	1

Step 1: Identify Pixels in V

The subsets share the boundary between columns 4 (rightmost of S_1) and 5 (leftmost of S_2).

S_1 **pixels with value 1:** $(3, 1), (4, 1), (3, 2), (2, 3), (3, 3)$ (using (x, y) coordinates from the grid).

S_2 **pixels with value 1:** $(7, 0), (8, 0), (6, 1), (5, 2), (6, 2)$.

The closest cross-boundary pair in V is $p = (4, 1) \in S_1$ and $q = (5, 2) \in S_2$.

Step 2: Test Each Adjacency Type

(a) 4-adjacency: Two subsets are 4-adjacent if $\exists p \in S_1, q \in S_2$ with values in V that are 4-neighbors (share an edge). The nearest pair is $(4, 1)$ and $(5, 2)$, which differ in *both* coordinates — they are *not* 4-neighbors. No other cross-boundary pair of pixels in V is closer.

$\Rightarrow S_1$ and S_2 are **NOT 4-adjacent**.

(b) 8-adjacency: $(4, 1)$ and $(5, 2)$ differ by 1 in each coordinate, so they are diagonal (8-)neighbors. Both have value $1 \in V$.

$\Rightarrow S_1$ and S_2 **ARE 8-adjacent**.

(c) m-adjacency: Two pixels p, q are m-adjacent if they are 8-adjacent *and* $N_4(p) \cap N_4(q) \cap V = \emptyset$ (no shared 4-neighbor has a value in V). The shared 4-neighbors of $(4, 1)$ and $(5, 2)$ are $(4, 2)$ and $(5, 1)$:

$$(4, 2) = 0 \notin V, \quad (5, 1) = 0 \notin V.$$

Since the intersection is empty, the m-adjacency condition is satisfied.

$\Rightarrow S_1$ and S_2 **ARE m-adjacent**.

Results

For $V = \{1\}$:

- (a) S_1 and S_2 are **not** 4-adjacent.
- (b) S_1 and S_2 **are** 8-adjacent (via $(4, 1) \leftrightarrow (5, 2)$).
- (c) S_1 and S_2 **are** m-adjacent (via $(4, 1) \leftrightarrow (5, 2)$, since $N_4(p) \cap N_4(q) \cap V = \emptyset$).

Problem 3: Shortest Paths (25 pts)

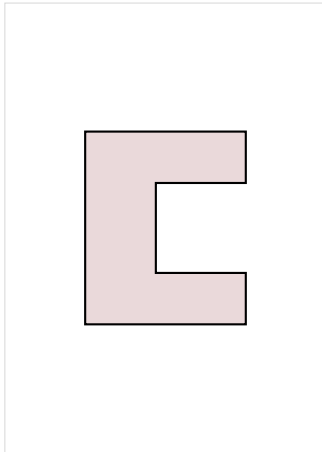
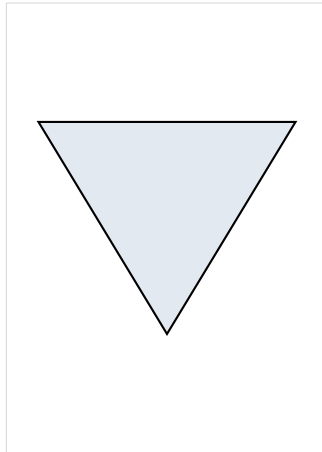
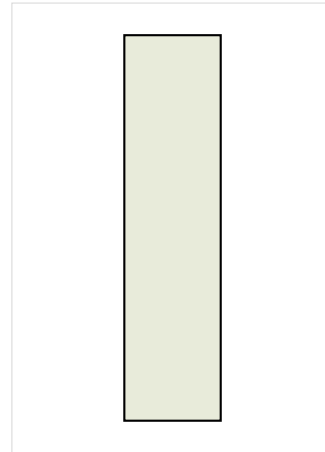
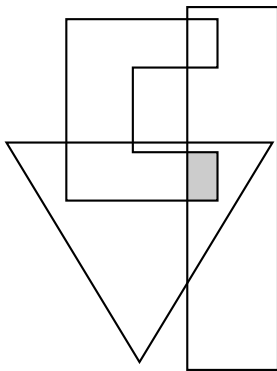
Consider the image segment shown below.

- (a) Let $V = \{0, 1\}$ and compute the lengths of the shortest 4-, 8-, and m-path between p and q . If a particular path does not exist between these two points, explain why.
- (b) Repeat for $V = \{1, 2\}$.

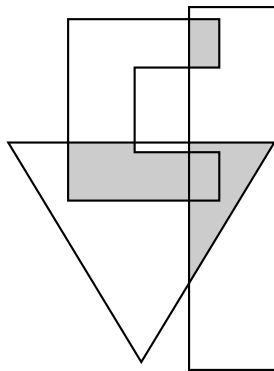
			(q)	
	3	1	2	1
	2	0	0	2
	1	2	1	1
(p)	1	0	1	2

Problem 4: Set Expressions (25 pts)

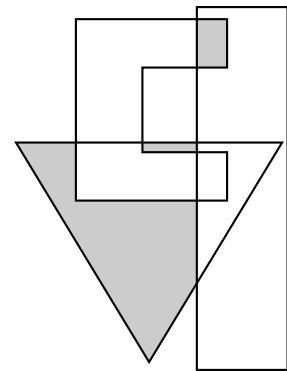
Give expressions for the sets shown shaded in the following figure in terms of sets A , B , and C . The shaded areas in each figure constitute one set, so give one expression for each of the three figures.

 A  B  C 

(i)



(ii)



(iii)